

HR ANALYTICS DASHBOARD

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Project Name: HR Analytics Dashboard

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Objective

The objective of the HR Analytics Dashboard is to analyze employee data and transform it into meaningful insights that help the Human Resources (HR) department make informed decisions. The dashboard provides an interactive view of key HR metrics such as employee count, attrition, salary, job satisfaction, work-life balance, performance, and promotions. It enables HR managers to identify workforce trends, improve employee retention, optimize resource planning, and enhance overall organizational performance through data-driven decision-making.

Exploratory Analysis

- Total number of employees
- Attrition count
- Attrition rate
- Department-wise employee
- Average monthly income
- Average years at company
- Job satisfaction levels

Data Visualization

- The HR Analytics Dashboard uses interactive visualizations such as KPI cards, bar charts, pie charts, donut charts, column charts, and slicers to present employee data in a clear and meaningful way.

- These visualizations allow HR managers to quickly understand workforce performance, compare departments, monitor attrition, and filter data based on different categories such as department, job level, and overtime status.

Performance Analysis

- Employee Attrition
- Department Performance
- Salary Distribution
- Employee Satisfaction

Dashboard Development

- Building an interactive dashboard using Microsoft Power BI and Excel.
- Using Power Query for data cleaning and DAX for calculated measures.

Problem Statement

Managing employee information manually can be time-consuming and may not provide meaningful insights for decision-making. HR departments often face challenges in tracking employee attrition, performance, salary trends, and satisfaction levels.

The HR Analytics Dashboard addresses these challenges by providing a centralized, interactive platform that transforms raw employee data into visual reports, enabling faster, more accurate, and data-driven HR decisions.

The project addresses the following challenges:

1. Cleaning and preparing employee data for analysis.
2. Identifying employee attrition trends across departments.
3. Monitoring employee performance and productivity.
4. Analyzing salary distribution and compensation patterns.
5. Evaluating job satisfaction and work-life balance. Presenting the results through an interactive dashboard.

Data Description

Source

The dataset was obtained from **Kaggle** and contains employee information for HR analytics. It includes various employee attributes such as age, department, job level, salary, years at the company, performance rating, job satisfaction, work-life balance, overtime, promotion history, and attrition.

1. Data Model Overview

The dataset follows a structured tabular format where each record represents an AI response evaluation containing information about the knowledge provided, user question, correct answer, hallucinated answer, response quality, risk level, and trust score.

Entities and Attributes

Attribute Name	Description
Employee ID	Unique identifier assigned to each employee.
Age	Age of the employee.
Department	Department where the employee works (e.g., HR, Sales, IT).
Job Level	Job designation or level of the employee.
Years at Company	Total years the employee has worked in the company.
Monthly Income	Monthly salary of the employee.

2. Data Relationships

The HR Analytics dataset follows a structured relationship between employee information and HR performance metrics. Each employee record contains

personal, job, and performance-related details that are connected to support workforce analysis.

- Each record represents one employee in the organization.
- Each employee belongs to one department.
- Each employee is assigned a specific job level.
- Each employee has a monthly income associated with their job role.
- Each employee has a recorded number of years at the company.
- Each employee has a job satisfaction rating.
- Each employee has a work-life balance rating.
- Each employee has a performance rating.
- Each employee has an overtime status (Yes/No).
- Each employee has a promotion status for the last five years.

Approach

1. Data Import

Software Used

- Microsoft Power BI Desktop
- Power Query
- Microsoft Excel

Process

- Collected the employee dataset from an Excel file.
- Imported the dataset into Microsoft Power BI Desktop.
- Reviewed all columns and verified the dataset structure.
- Checked and corrected data types for each column.
- Loaded the dataset into Power BI for further analysis.
- Prepared the data for cleaning, transformation, and visualization.

2. Data Cleaning

- Checked and removed missing or blank values from the dataset.

- Removed duplicate employee records to ensure data accuracy.
- Corrected and verified data types for all columns.
- Standardized categorical values for consistency.

3. Data Analysis and Visualization

The HR Analytics Dashboard was created using **Microsoft Power BI** and **Microsoft Excel** to analyze employee data and present meaningful insights through interactive visualizations.

The dashboard includes:

- **KPI Cards** displaying Total Employees, Attrition Count, Attrition Rate, Average Monthly Income, and Average Years at Company.
- **Bar Chart** showing Department-wise Employee Distribution.
- **Pie/Donut Chart** displaying Gender Distribution and Job Satisfaction Levels.
- **Clustered Column Chart** comparing Monthly Income across Departments.
- **Column Chart** showing Performance Ratings and Training Hours.
- **Pie Chart** displaying Overtime Status and Promotion Status.
- **Interactive Slicers** for filtering the dashboard by Department, Gender, Job Level, Attrition, and Overtime

Project Result

1. Data Collection and Preparation

The HR analytics dashboard dataset was imported from CSV files downloaded from Kaggle.

The dataset was cleaned using Power Query to remove inconsistencies and prepare it for analysis.

2. KPI Dashboard Displays:

- Total Employees
- Attrition Count
- Attrition Rate
- Average Monthly Income
- Average Years at Company

Result:

Provides a quick overview of overall HR Analytics Dashboard

3. Donut Chart:

Attrition Analysis

Result:

Employee Attrition based on **OverTime (Yes/No)**.

4 Bar Chart:

Department Wise Employees

Result:

The bar chart displays the total number of employees in each department.

5. Line Chart:

Years at Company

Result:

The line chart shows the relationship between employees' years at the company and average salary.

6. Area Chart:

Performance Rating

Result:

The area chart shows the distribution of employees based on their performance ratings.

7. Gauge Chart:

Job Satisfaction

Result:

The gauge chart displays the overall job satisfaction score of employees.

8. Waterfall Chart:

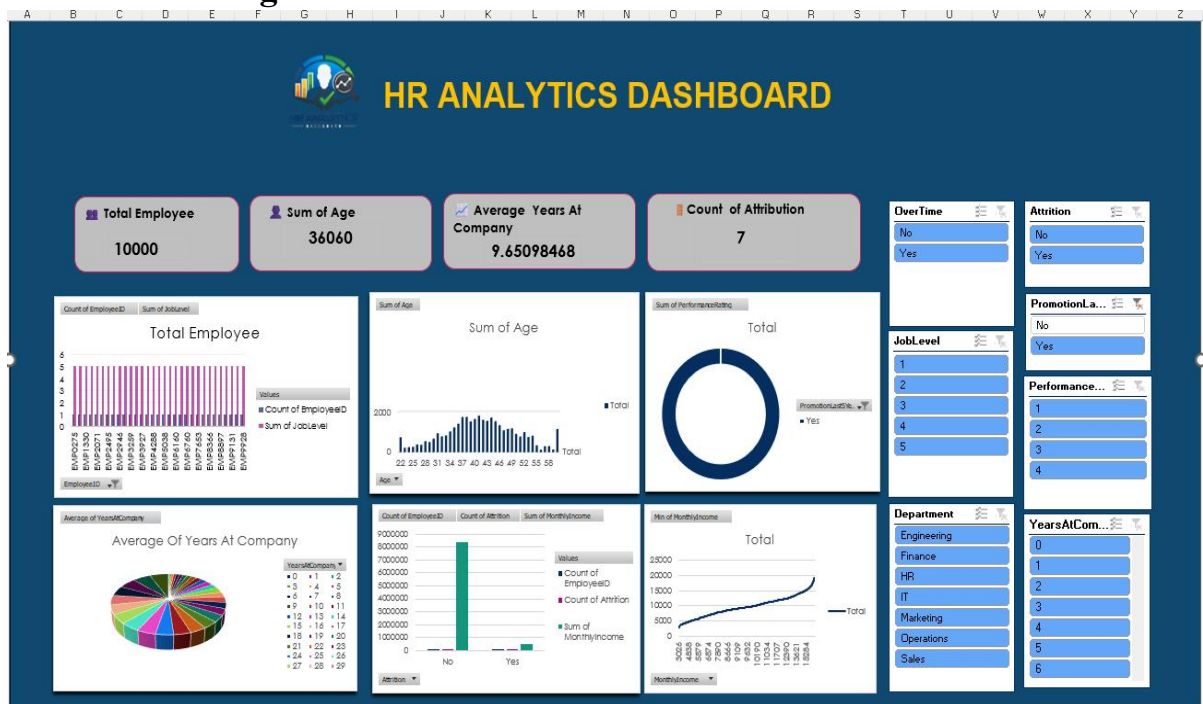
Salary by Job Level

Result:

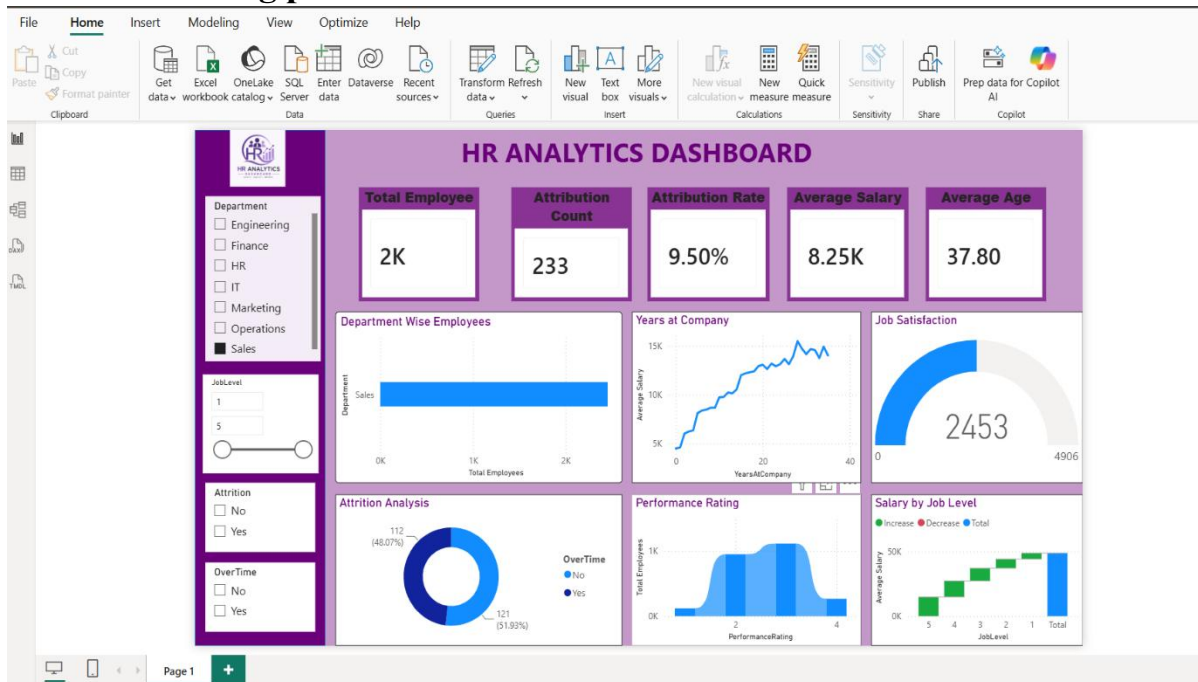
The waterfall chart illustrates how average salary changes across different job levels.

Project result

Dashboard using excel:



Dashboard using powerBi



Conclusion

The **HR Analytics Dashboard** provides an interactive and comprehensive overview of employee data using **Microsoft Power BI**. It transforms raw HR data into meaningful insights by analyzing employee count, attrition, salary, job satisfaction, performance, and years at the company. The dashboard enables HR managers to monitor key workforce metrics, identify trends, and make informed, data-driven decisions.

The dashboard demonstrates the practical application of Business Intelligence techniques such as:

- Data Collection
- Data Cleaning
- Data Modeling
- Data Visualization
- KPI Development
- DAX Analysis
- Interactive Dashboard
- Dashboard design